

Adrian Lozada

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

Expected Graduation: 12/2026

M.S. Computational Science and Engineering

3.33/4.0

- Relevant Coursework: **High Performance Parallel Computing, Numerical Linear Algebra, Modeling and Simulation**

University of South Florida (USF), Tampa, FL

08/2021 - 05/2025

B.S. Computer Engineering, summa cum laude

3.9/4.0

- Honors: FLIT-GAP Scholar, Bright Futures Scholar
- Relevant Coursework: Analysis and Design of Algorithms, Data Structures, Social Networks, Computer Architecture

PROJECTS

High-Performance Parallel Computing

- Parallelized dense matrix kernels using **OpenMP** work-sharing and loop-collapse strategies in **C++**, achieving near-linear **16x** speedup at 16 threads on GT's PACE/ICE cluster.
- Accelerated a heat equation PDE solver on an **NVIDIA H100** via **CUDA** shared memory tiling and **OpenMP** GPU offloading, achieving up to **83.9x** and **17x** speedup respectively over single-threaded CPU baselines.
- Implemented **MPI**-parallel Monte Carlo simulation scaling to 64 processes, achieving **99.7%** parallel efficiency at 16 processes and **16x** strong scaling speedup on 1-billion-sample workloads.
- Applied **Roofline** performance modeling to classify SpMV, stencil, and 3D-FFT kernels as memory-bound on Intel and AMD architectures, identifying bandwidth bottlenecks and cache optimization strategies.

Autonomous Ground Vehicle with Remote Monitoring System

- Led a team of 7 to design and build an autonomous vehicle on **Raspberry Pi**, mentoring peers in **Linux** development and conducting code reviews to deliver a working prototype.
- Designed and deployed a **Flask** backend with **SQLAlchemy** to expose REST endpoints for robot status, logs, and commands.

Graph Neural Network-Based Fraud Detection

- Developed a Graph Neural Network in **PyTorch** to classify fraudulent financial transactions, achieving **91%** accuracy on real-world datasets.
- Preprocessed graph data with **NetworkX**, extracting structural features (centrality, clustering coefficients) to improve model performance.

EXPERIENCE

Software Engineer Intern

05/2025 - 08/2025

Orion Defense Solutions L.L.C.

(Remote)

- Built and deployed a **Python/FastAPI** automation platform integrating **LangChain** and **SQL**-backed internal systems, reducing project cycle time by **~8 hours**.
- Containerized platform microservices using **Docker** and **Kubernetes**, enabling rolling updates and eliminating **~7 hours** of manual setup per week.

Undergraduate Researcher

01/2024 - 05/2025

Reality, Autonomy, Robot Experience (RARE) Lab, USF

Tampa, FL

- Published two first-author papers at ACM/IEEE HRI 2025 (**25% acceptance rate**) on autonomous fog-screen communication systems.
- Developed ROS nodes in **C++** to synchronize Fetch robot behavior, achieving reliable distributed actuation.

SKILLS

Languages: Python, C/C++, SQL, Bash/Zsh, JavaScript

Systems/Areas: Data Structures & Algorithms, Distributed Systems, HPC, Unix/Linux

Tools: Git, Docker, Kubernetes, ROS

Parallel/HPC: MPI, OpenMP, CUDA

Libraries: PyTorch, NumPy, Pandas, FastAPI, Flask, SQLAlchemy, LangChain, NetworkX

AWARDS

UR2PhD Technical Conference Travel Award, Computing Research Association (CRA) (\$2000)

03/2025

Tampa Conference Presentation Grant Program (CPGP) (\$700)

03/2025

NSF Distributed Research Experiences for Undergraduates (DREU) (\$7000)

05/2024 - 08/2024

Florida Bright Futures Scholarship (100% of Tuition)

08/2021 - 05/2025